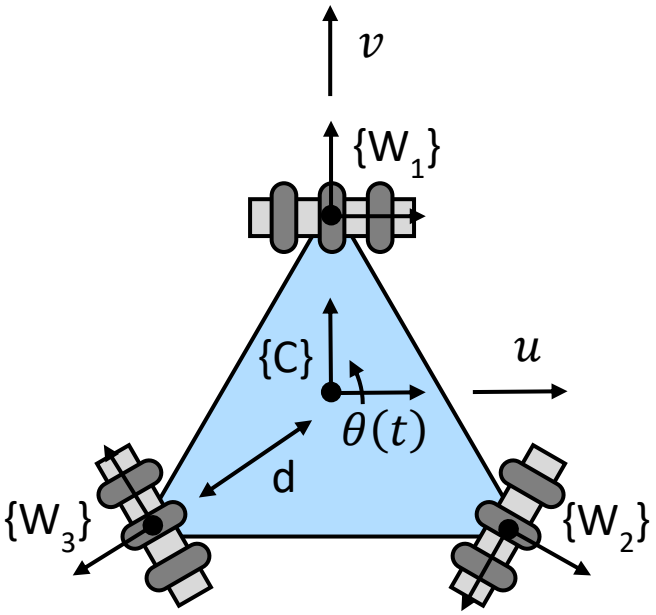
ENN 581 Semester 2
Week 10 – Mobile robot kinematics

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1. Consider the following robot with three omnidirectional wheels. Each of these can be driven independently. Their angular velocities are given as $\dot{\varphi}_1$, $\dot{\varphi}_2$ and $\dot{\varphi}_3$. The rollers are oriented perpendicular to the driving direction of each wheel.

   The geometry of the robot is an equilateral triangle. The distance from the centre to each vertex is given as d.



   Derive a kinematic expression that relates the angular velocities of the wheels ($\dot{\varphi}_1$, $\dot{\varphi}_2$ and $\dot{\varphi}_3$) to the velocity of the robot chassis frame ($u, v, \dot{\theta}$).

2. If d = 10 cm, write a python script that analyses the motion of the robot in an inertial frame. Change the angular velocities of the wheels and see how the robot moves. Plot both the velocities, positions, and orientations of the robot. You may use simple Euler integration to determine positions (e.g., $x(t + dt) = x(t) + \dot{x}(t)dt$).

3. For the differential drive robot that we discussed in the lecture, derive the position of the instantaneous centre of rotation, if the rear wheels are rotating with velocities $\dot{\varphi}_1$ and $\dot{\varphi}_2$.

4. Derive the kinematic equations for a bicycle type robot. The front wheel is steered and not driven. The rear wheel is fixed and driven. Is this system holonomic or non-holonomic?